

Chemical Engineering Student Experiences

ChE Retreat August 17, 2022



Breadth of ChE Curriculum

Core Courses

Thermodynamics-Based Courses

CHE 201 – Introduction to thermodynamics
CHE 301 – Chemical Engineering Thermodynamics
CHE 321 – Chemical Reaction Engineering

Process-Oriented Courses

CHE 210 – Material & Energy Balances
CHE 341 – Chemical Process Control
CHE 396 – Senior Design I (Equipment Design)
CHE 397 – Senior Design II (Design teams)

Numerical Analysis

CHE 205- Numerical Methods in ChE

Engineering Education & Development

ENGR100 -Engineering Orientation (Engineering Success Program,
Guaranteed Paid Internship Program)
CHE 499- Professional Development Seminar

Transport-Based Courses

CHE 311 – Transport Phenomena I (Fluid Mechanics)
CHE 312 – Transport Phenomena II (Heat & Mass Transfer)
CHE 313 – Transport Phenomena III (Separations)

Unit Operations Laboratory

CHE 381 – Chem Eng Lab I
CHE 382 – Chem Eng Lab II

Molecular Engineering Courses

CHE 230 Molecular Engineering in ChE
CHE 330 Polymer Science

CHE 392:
Undergraduate
Research

Specializations

Biochemical/Pharmaceuticals

CHE 422 Biochemical Engineering
CHE 425 Nanotech for Pharmaceuticals
CHE 494 Biopharma Process Eng.

Energy

CHE 451 Renewable Energy Technology
CHE 453 Electrochemistry
CHE 494 Clean Energy & the Electric Grid

Environmental

CHE 450 Air Pollution Engineering
CHE 453 Electrochemistry
(CHE 150 Climate Engineering)

Process Simulation

CHE 433 Process Simulation w/Aspen+

Entrepreneurship

CHE 427 Entrepreneurship in Engineering

Molecular Engineering

CHE 454 Mol. & Macromol Engineering
CHE 455 Nanoscale Systems in ChE
CHE 457 Colloidal & Interfacial Phenom

Undergraduate Curriculum Improvement

- Strengthen core courses
 - Student learning

New Core Course?

CHE 100 Introduction to Chemical Engineering

- New 400-level courses
 - Existing specializations
 - Additional areas- future topics trending in CHE, faculty specialties
- Other ideas?

ChE Extracurricular Activities

- Internships
 - Student-led/Engineering Career Center
 - CHE 391
- AIChE Student Chapter
 - ChemE Car
 - AIChE National meeting, North-Central Regional Conference, Chicago Section meeting (poster session)
- ChE Industry Day & ChemE Forum
- ACCESS Club (BB)
- Summer webinars (MC)
- Other student organizations (SWE, SHPE, etc)

Extracurricular Activity Improvement

Ideas?